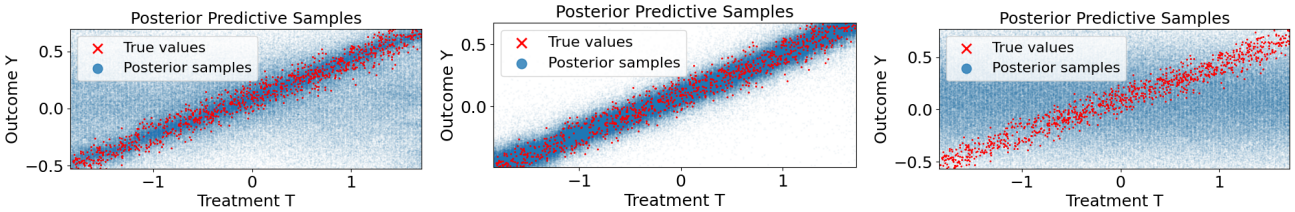

Figure 1. Here, we qualitatively demonstrate what happens when providing a Causal Foundation Model (CFM) with incorrect graph information in the unidentifiable bivariate case of a linear mechanism and Gaussian noise. Since the data does not provide any evidence regarding the causal direction, the model outputs the "wrong" posterior, while performing correct inference under the wrong assumption: More specifically, this plot shows posterior predictive samples in the two-node case where T causes Y ($T \to Y$): i.e., $Y = 0.25 \cdot T + \epsilon$, for $\epsilon \sim \mathcal{N}(0, 0.1)$. This makes the causal direction is not identifiable from observational data. **Left**: Without providing the model Q_θ with any information regarding the causal direction, it outputs a mixture distribution $Q_\theta(Y|\text{do}(T), T \overset{?}{\longleftrightarrow} Y) = 0.5 \cdot P(Y|\text{do}(T), Y \to T) + 0.5 \cdot P(Y|\text{do}(T), T \to Y)$. **Middle**: When conditioned on the right causal graph, the output aligns with the correct causal effect: $Q_\theta(Y|\text{do}(T), T \to Y) = P(Y|\text{do}(T), T \to Y)$. **Right:** When provided with the wrong causal information $T \leftarrow Y$, the model makes the prediction $Q_\theta(Y|\text{do}(T), T \leftarrow Y) = P(Y|\text{do}(T), T \leftarrow Y)$, which is the correct thing to do under the wrong assumption $T \leftarrow Y$.